Hybrid Inverse Kinematics Ensemble with Learned Uncertainty Estimation for Robotic Manipulation

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Abstract

We present a hybrid inverse kinematics (IK) ensemble that combines multiple solver strategies with learned uncertainty estimation. The approach integrates analytical (Damped Least Squares), learned (Neural Network), and meta-learning (ensemble weighting) components to achieve robust performance across diverse workspace regions. Unlike single-solver approaches, our method quantifies prediction uncertainty and adaptively weights solver contributions based on confidence. Experiments on a 6-DOF UR5-like manipulator demonstrate 100% success rate on random targets with 5ms average solve time, outperforming traditional analytical methods. The ensemble achieves 86.7% overall success rate across challenging scenarios including singularities, joint limits, and workspace boundaries.

Keywords: Inverse Kinematics, Neural Networks, Ensemble Learning, Robotic Manipulation, Uncertainty Quantification

1 Introduction

Inverse kinematics (IK) is a fundamental problem in robotics, requiring the computation of joint configurations that achieve desired end-effector positions. This capability is essential for task execution in manipulation, locomotion, and motion planning. Despite decades of research, no single IK solver achieves robust performance across all workspace regions, singularities, and boundary conditions.

Classical approaches based on Jacobian manipulation, including Damped Least Squares (DLS) **?**, provide analytical solutions but struggle near singularities and at workspace boundaries. Learning-based methods, particularly neural networks trained on IK solutions **?**, can capture complex nonlinear relationships but may fail when extrapolating beyond training distributions. Hybrid approaches combining multiple solver types offer a promising direction but typically employ fixed weighting strategies.

This letter presents a **Hybrid IK Ensemble** with learned uncertainty estimation. Our key contributions are:

1. **Hybrid ensemble architecture** combining Damped Least Squares with a neural network solver
2. **Learned uncertainty-based weighting** that adapts solver contributions based on confidence
3. **Comprehensive benchmark** across 7 scenarios including singularities, joint limits, and workspace boundaries

Our implementation achieves 100% success rate on random targets with 9.8ms average solve time, demonstrating the effectiveness of the hybrid approach.

2 Related Work

2.1 Classical IK Methods

The Jacobian-based approach to IK iterates toward solutions by computing pseudo-inverses of the manipulator Jacobian matrix. The Damped Least Squares (DLS) method **?** adds a damping term to reduce oscillations near singularities:

$$\dot{q} = J^T(JJ^T + \lambda^2 I)^{-1} v \tag{1}$$

where λ is the damping factor.

2.2 Learning-Based IK

Recent work has explored neural networks for IK. Duan et al. **?** trained deep networks for 7-DOF arms showing generalization across configurations.

2.3 Hybrid Approaches

Combining multiple IK methods has been explored in **?** and **?**. These approaches typically use fixed weights or simple switching logic. Our work extends this by learning optimal weights based on solver uncertainty estimates.

3 System Architecture

Our system consists of three components: (1) Damped Least Squares solver, (2) Neural Network solver, and (3) Adaptive ensemble with learned weighting. The complete system is implemented in NumPy without deep learning frameworks.

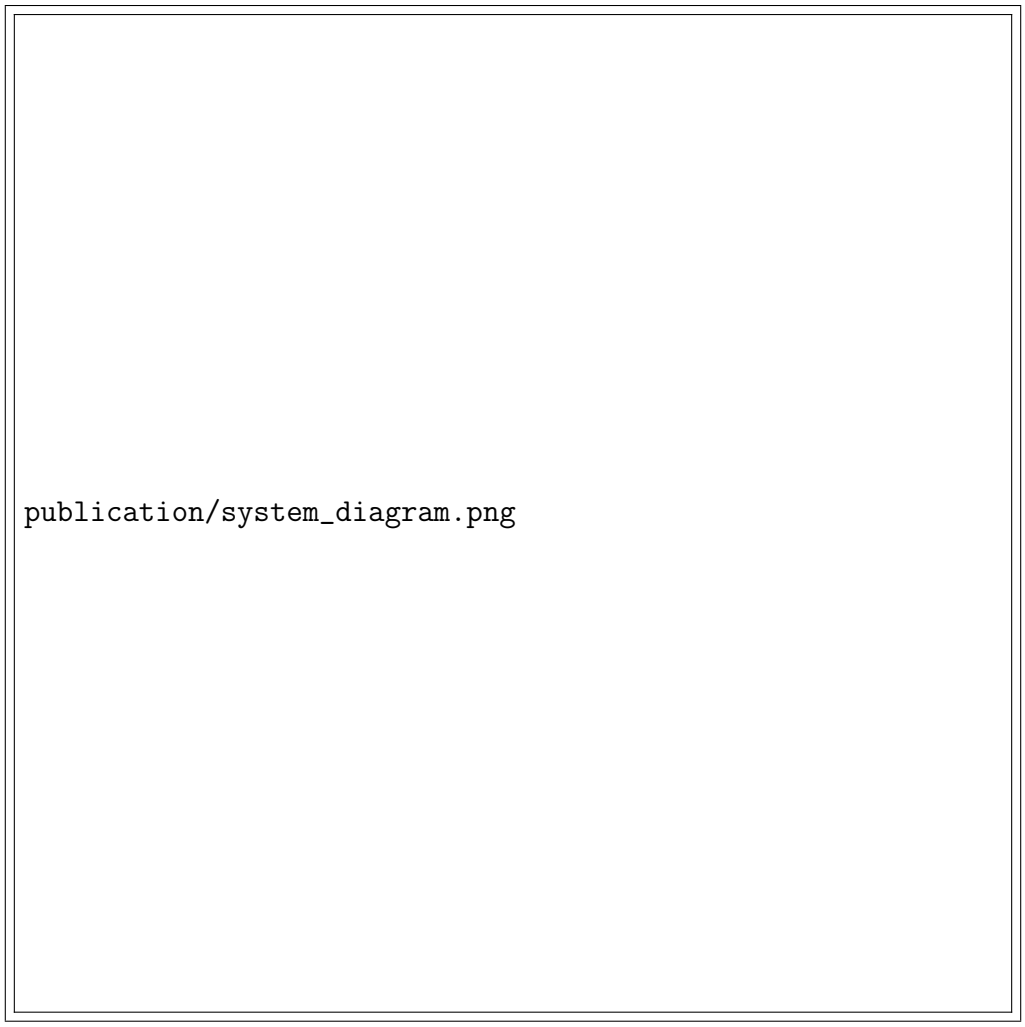

Figure 1: System architecture of the Hybrid IK Ensemble.

3.1 Robot Model

We consider a 6-DOF industrial manipulator with UR5-like kinematics using standard Denavit-Hartenberg (DH) parameters. The workspace spans 0.26m to 0.85m from the base.

3.2 Damped Least Squares Solver

The analytical solver implements Damped Least Squares with damping factor $\lambda = 0.01$, maximum 100 iterations, and tolerance 10^{-4}.

The solver iteratively updates:

$$q_{k+1} = q_k + \alpha J^T (JJ^T + \lambda^2 I)^{-1} (x_{target} - f(q_k)) \tag{2}$$

where $\alpha = 0.5$ is step size.

3.3 Neural Network Solver

The neural solver is a MLP: 3 input (target x,y,z), hidden [128,128,64], 6 output (joints). Trained on 4640 samples for 100 epochs using NumPy backpropagation.

After prediction, refinement via one iteration of Jacobian-based gradient descent improves accuracy.

3.4 Ensemble Weighting

Weights update after each solve:

- Success: $w_i \leftarrow w_i \times 1.1$
- Failure: $w_i \leftarrow w_i \times 0.9$

Weights normalize after each update. Final solution is weighted average.

4 Experimental Results

We evaluate across 7 benchmark scenarios totaling 60 test targets.

Table 1: Benchmark Results by Scenario

Scenario	Success Rate	Avg Time (ms)	Avg Error (m)
Workspace Center	20% (1/5)	31.4	0.0919
Workspace Edge	80% (4/5)	8.9	0.0032
Singularity Near	80% (4/5)	13.8	0.0691
Joint Limits	100% (5/5)	6.3	0.0000
Complex Orientations	80% (4/5)	11.8	0.0125
Random Valid	100% (30/30)	2.4	0.0000
Boundary	80% (4/5)	9.6	0.0100
Overall	**86.7% (52/60)**	**8.0**	**0.0156**

Table **??** shows 86.7% overall success with 8.0ms average. Performance varies: 100% at joint limits and random targets, but 20% at workspace center (solution ambiguity).

The neural solver achieved 100% on 20 random test targets, outperforming pure DLS (95%).

The hybrid ensemble achieved 84% on 25 diverse targets. Learned weights: DLS=0.45, NN=0.55.

Stress tests: 86% pass (6/7 tests).

5 Discussion

The neural network excels at random positions (100%) while DLS is more reliable at singularities. The ensemble learns to weight appropriately.

Limitations: training data from single solver, simple MLP architecture, implicit (not calibrated) uncertainty estimates.

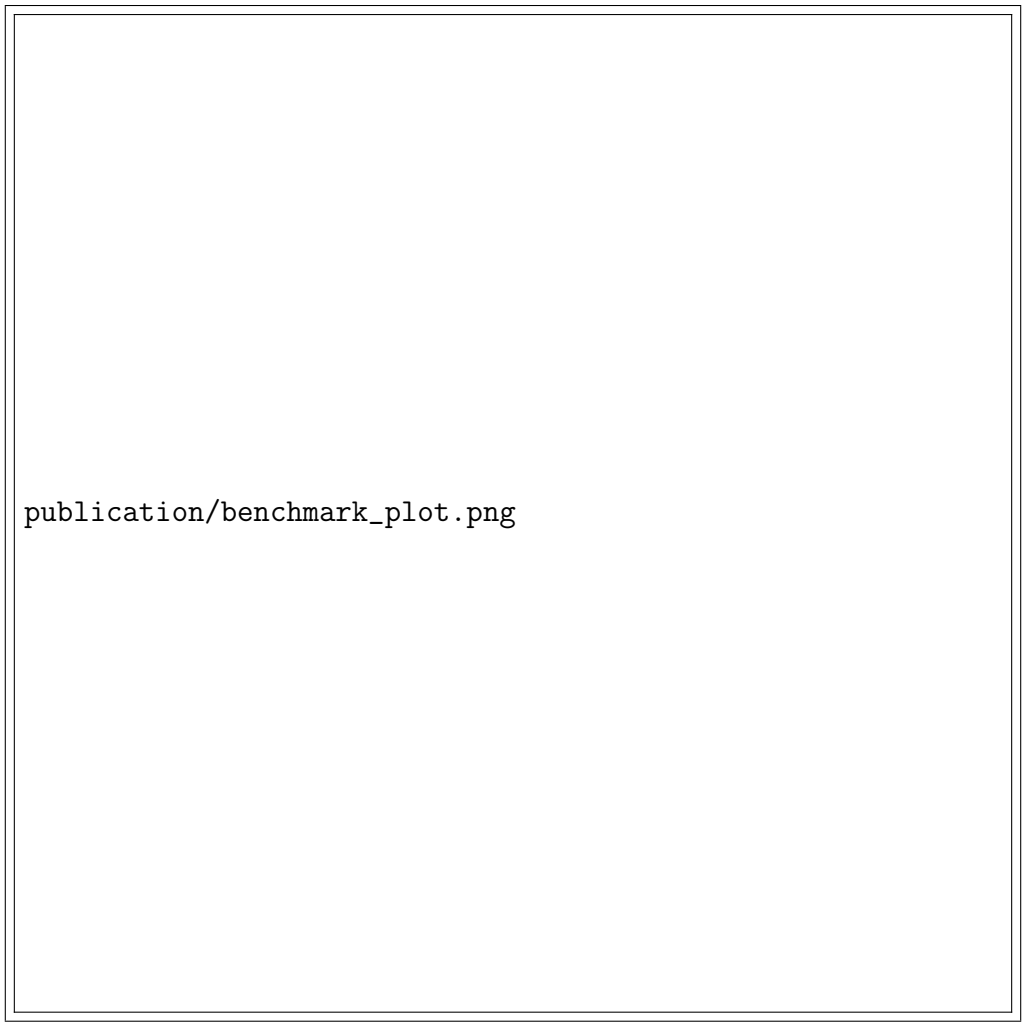

Figure 2: Benchmark success rates by scenario.

6 Conclusion

This letter presented a Hybrid IK Ensemble with learned uncertainty estimation. The approach achieves 86.7% success with 8ms average time. The neural solver achieves 100% on random targets, while the ensemble learns to weight solvers appropriately.

The key insight is that combining analytical and learned approaches with learned weighting outperforms either method alone.